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Title: Exertional Heat Stroke: An Evidence Based Approach to Clinical Assessment and Management

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Running Title: Exertional Heat Stroke Treatment

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New Findings: 1. What is the topic of this review?

•The treatment of exertional heat stress, from initial field care through the return-to-activity decision, is reviewed.

2. What advances does it highlight?

• Clinical assessment during field care using AVPU and vital signs to gauge recovery
•Approaches to field cooling and end of active cooling
•Shared clinical decision making for return to activity recommendations

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Exertional Heat Stroke: An Evidence Based Approach to Clinical Assessment and Management

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Abstract

Exertional heat stroke (EHS) is a potentially fatal condition characterized by central nervous system dysfunction and body temperature often but not always >40°C that occurs in the context of physical work in warm or hot environments. In this paper, we review the continuum of care, from initial recognition and field care to transport and hospital care, and finally return to duty considerations. Morbidity and mortality can be greatly reduced if not eliminated with prompt recognition and aggressive cooling. If medical personnel are not present at point of collapse during or immediately following exercise, EHS should be the presumptive diagnosis until a formal diagnosis can be determined by qualified medical staff. EHS is the rare medical situation where initial treatment (cooling) takes precedence over transport to a medical facility, where advanced medical care may be required for severe EHS casualties. Recovery from EHS and return to activity is usually straightforward and unremarkable provided the casualty is rapidly cooled at time of collapse and adequate time is allowed for body healing. However, evidence-based data to guide return to activity following EHS is limited. Current research suggests that most individuals recover completely within a few weeks

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Introduction

Exertional heat stroke (EHS) is the most severe form of exercise-related heat illness and may result in severe disability or death if not promptly identified and managed appropriately (Epstein & Yanovich, 2019). EHS mortality rates display considerable variability dependent upon the context of the heat stress and the availability of rapid body cooling. On one extreme are data from mass participation sporting events, in which medical personnel use on-site cold water immersion to cool all suspected heat illness casualties. During an 18-year period at the 11.4k,/7.1 mile Falmouth Road Race, there were 274 EHS cases with no fatalities (Demartini *et al.*, 2015). On the other extreme are data from the occupational sector in which 23 of 27 (85%) EHS casualties investigated by the US Occupational Safety and Health Administration resulted in death (Tustin *et al.*, 2018). This report makes no mention of on-site cooling interventions and considering the extremely high mortality rate, it is reasonable to assume that on-site cooling was not available or performed (Heled *et al.*, 2004; Rav-Acha *et al.*, 2004). Other studies report a range of EHS mortality rates with values of 5-10% typically reported as representative of the literature (Shapiro & Seidman, 1990; Leon & Bouchama, 2015; Epstein & Yanovich, 2019). While there is a robust body of literature concerning the risk factors that predispose an individual to EHS (Carter *et al.*, 2005; Nelson *et al.*, 2018; Alele *et al.*, 2020; Kerr *et al.*, 2020; Westwood *et al.*, 2020), consistent observations identify that limiting EHS morbidity and mortality is largely dependent on prompt, and aggressive whole-body cooling (Belval *et al.*, 2018; Roberts *et al.*, 2021).

The emphasis on prompt and aggressive cooling is to limit the time-temperature area under the curve, or the 'dose' of heat experienced by the casualty. In rodent models of classic and EHS mortality increased with increased time- temperature product (Hubbard *et al.*, 1976). In a report of classic heat stroke (CHS), which results from passive exposure to extreme heat, delayed cooling (core temp >38.9°C after an hour in the emergency department) had a mortality rate more than twice that of those who were cooled to <38.9°C within an hour (Vicario *et al.*, 1986). Similarly delayed cooling in EHS has been associated with an increase in mortality (Heled *et al.*, 2004; Rav-Acha *et al.*, 2004). Accordingly, EHS is a well-recognized medical emergency with the potential for tissue destruction and cell death in the absence of prompt recognition and management. Rapid recognition of EHS with onsite cooling capabilities has been clearly demonstrated to improve clinical outcomes (Costrini, 1990; Demartini *et al.*, 2015). Comparable to cardiac arrest and brain stroke, time is tissue, and a planned and systematic response can be lifesaving. Surviving and recovering from EHS requires a multidisciplinary team effort to execute a chain of survival that integrates prevention, prehospital care, emergency transport, advanced clinical care, and a progressive return to activity (See Figure 1) (Bouchama *et al.*, 2022). Coordinated care should be integrated into a site-specific emergency action plan. When treated appropriately, especially when initially recognized at point of injury, morbidity and mortality can be greatly limited or even eliminated. This evidence-based review will discuss initial recognition, pre-hospital and hospital care, recovery and return to activity considerations. The cornerstone of EHS management, prevention, is discussed in a separate review in this issue.

Insert Figure 1 about here

Prehospital Care

Rapid Recognition

EHS is defined as a syndrome of central nervous system dysfunction, with body core temperature typically but not always greater than 40ºC (Leon & Bouchama, 2015; Epstein & Yanovich, 2019; Bouchama *et al.*, 2022). In most cases, identifying a suspected EHS casualty occurs when the individual collapses. Signs and symptoms prior to physical collapse may include ataxia, gait instability, personality changes, irrational behavior, and/or altered mental status. When any of these signs or symptoms are observed in the context of physical work in warm-to-hot environments, EHS should be the presumptive diagnosis pending medical evaluation to rule out other possibilities. The initial signs and symptoms of exertional heat stroke are similar to other exercise related conditions like heat exhaustion, exercise associated hyponatremia, and exercise associated postural hypotension (exercise associated collapse). Early recognition and immediate cooling improve outcomes by reducing the area (degree-minutes) under the body temperature vs time curve (Belval *et al.*, 2018). Several high-profile cases have demonstrated the catastrophic consequences associated with delayed diagnoses (Grundstein *et al.*, 2017; Independent Evaluation, 2018). EHS can also occur in cool environments and should be considered as a potential cause of collapse in exercise settings, especially when accompanied by unexplained neurologic symptoms (Roberts, 2006).

Rapid Assessment

When body temperature rises above a safe level, the cellular changes associated with hyperthermia begin. Cytotoxicity and tissue damage are functions of both the degree and duration of hyperthermia (Vicario *et al.*, 1986; Bouchama *et al.*, 2005). Collapse from EHS does not coincide with the point that body temperature crosses this critical threshold, so hyperthermia is present for an undetermined time. The initial cell changes that occur with hyperthermia are reversible when the time in the hyperthermic range is minimized and treatment to begins before the heat related cell changes become irreversible. Figure 2 presents a hypothetical model of core temperature during an EHS event. Minimizing the time from collapse (red arrow) to initiation of whole-body cooling (yellow arrow) is associated with good clinical outcomes. The long term sequala of tissue hyperthermia are not known.

Insert Figure 2 about here

If medical personnel are available, core (rectal) temperature should be recorded as soon as possible, as an indicator of severity and to guide treatment; temperature measurement at other anatomical sites is not recommended (Ronneberg *et al.*, 2008; Ganio *et al.*, 2009). If collapse occurs during or immediately following the cessation of heavy physical work and a rectal temperature cannot be measured, EHS should be the presumptive diagnosis. Technologies are currently being tested that may allow real-time physiological monitoring of participants and for the identification of a EHS casualty prior to collapse (Buller *et al.*, 2021). Using a combination of tri-axial accelerometry and core temperature prediction of heart rate and skin temperature, retrospective analysis allowed for the detection of EHS casualties 10-15 minutes prior to collapse (Buller *et al.*, 2022). In the future, it may be possible for an individual participant to self-monitor their heat status using these technologies. However, as altered mental status frequently occurs in EHS casualties, developing remote monitoring is required to optimize safety and are an area of active research by the US Army (Buller *et al.*, 2018).

Making the diagnosis of exertional heat stroke integrates central nervous system (CNS), cardiovascular, and body temperature assessment (see Table 1). CNS changes associated with hyperthermia are the hallmark of EHS. CNS changes vary from subtle to profound and can be initially assessed using the AVPU scale commonly used in emergency care. AVPU is the acronym for alert, response to verbal stimulation, response to pain stimulation, and unconscious response to stimulation and can be used to estimate the Glasgow Coma Score (GCS) of 0 (unconscious and unresponsive) to 15 (normal response). Cranial nerve, ocular movements, and pupil reactivity and signs are other simple CNS assessments. Decorticate and decerebrate posturing can occur. CNS function is also reflected in behavioral changes that can vary from subtle personality changes to aggressive temporal-frontal lobe type changes or cognitive changes like missing assignments or appearing to be "lazy." Vomiting can be a sign of EHS from a central origin due to brain heating or a gastrointestinal origin due to slowed motility from an overheated and under perfused gut (Zhong *et al.*, 2021).

Insert Table 1 about here

Vital signs deviate from normal with rectal temperature elevation the most obvious marker of EHS. A rectal temperature greater than 40ºC is common in EHS casualties, and unconscious casualties are often greater than 42ºC (Epstein &

Yanovich, 2019). In unwitnessed situations, a victim with a rectal temperature less than 40ºC may still have EHS due to spontaneous cooling. Rectal temperature measurement is the accepted field method, and oral, aural, axillary, or temporal artery skin temperatures cannot be substituted for rectal measurements (Roberts, 1994; Ronneberg *et al.*, 2008; Ganio *et al.*, 2009). Tachycardia, hyperventilation, and hypotension are physiological responses to hyperthermia and skin vasodilatation. Physical exam of the skin can be paradoxical. In CHS the skin may be hot and dry due to the absence of sweating, while in EHS the skin is usually wet and warm or cool to touch due to the evaporation of sweat (Bouchama *et al.*, 2022). Skin color is pale and ashen in severe EHS and can be flushed or become flushed as skin blood flow returns to normal.

Rapid Cooling

Exertional heat stroke is one of the rare exceptions in which treatment (cooling) is prioritized over transport (Belval *et al.*, 2018). It is essential that mass participation sporting event organizers, athletic team medical personnel and military leaders adequately prepare for the possibility of EHS casualties when conditions warrant. American football at the high school (Kerr *et al.*, 2013) and collegiate (Yeargin *et al.*, 2019) levels has the highest rates of EHS in training and in competition. Road races of varying distances have also been associated with relatively high frequency or incidence rates of EHS (Demartini *et al.*, 2015; Breslow *et al.*, 2021; Roberts *et al.*, 2021). In the military setting, timed run and foot march events were responsible for 80% of all EHS casualties during a 5-year surveillance period (DeGroot *et al.*, 2022).

The primary focus of EHS treatment is whole body cooling (Vicario *et al.*, 1986; Bouchama & Knochel, 2002; Bouchama *et al.*, 2007; Belval *et al.*, 2018; Roberts *et al.*, 2021). Cooling should be started on site and continued during transportation to and in the hospital until the body temperature in in the normal range. On site, the patient should be moved to the coolest place immediately available and outer layers of clothing removed. There are several body cooling methods used in the field with good clinical outcomes. Cooling techniques are categorized into conductive and evaporative methods. Conductive cooling is based on establishing the largest possible temperature gradient between the core and the skin to accelerate heat transfer to the surrounding ice, cold water, cold towels, or cold blanket in contact with the skin (Wyndham *et al.*, 1959; Bouchama & Knochel, 2002; Bouchama *et al.*, 2007). There is no evidence favoring any cooling method over another in the therapy of classic heatstroke (Bouchama *et al.*, 2007). Immersion in cold (ice water) or cool (freely running tap) water are well documented methods for treating EHS in the field, although similar clinical outcomes can be obtained with rapidly rotating ice water towels, rotating ice water-soaked sheets, repeated dowsing with cold or cool water, and cool water spray or sponging with fans (in low humidity environments) (McDermott *et al.*, 2009b; Douma *et al.*, 2020). Cooling by evaporation is centered on the thermodynamic principle that the conversion of 1.7 ml of water from the skin to vapor will result in a heat loss of 1 kcal (Wyndham *et al.*, 1959; Bouchama *et al.*, 2007). The effectiveness of evaporative cooling depends on creating a high water vapor pressure gradient, which is achieved via water wetting of the body surface or covering the skin with wet gauze and providing continuous fanning. In high humidity environments evaporative cooling is less effective and other techniques should be considered. Table 2 outline techniques and cooling rates commonly used in the field.

Insert Table 2 about here

On-site cooling prevents treatment delays and cooling interruptions associated with transportation to medical facilities and emergency department evaluation protocols for encephalopathy; cooling rates above 0.08ºC/min are best for survival without medical complications in casualties identified early in the course of EHS (McDermott *et al.*, 2009a; Filep *et al.*, 2020). Initiating whole-body cooling as soon as possible is essential; the method used will depend on site assets and limitations, water and ice access, patient size and body type, and the incidence rate of EHS at the site. When a rectal thermometer is not available to measure the temperature in a setting where EHS is likely to occur and other symptom and signs are consistent with EHS, whole body cooling should be started by any means available until the temperature can be measured on-site, in an ambulance, or at the ED.

Cold water immersion is the preferred method for the treatment of EHS. While DeMartini et al document no deaths using partial body tub immersion with average cooling rates above 0.15ºC/min, the Falmouth Road Race has had no EHS deaths since its inception including the years before onsite tub cooling (Demartini *et al.*, 2015). There are no published cases of EHS deaths in road racing when the affected runners are identified early and cooled quickly with a variety of methods (Filep *et al.*, 2020). However, in some settings this may not be practical and other methods have been recommended.

Tarp-assisted Cooling, or TACO method, provides an improvised tub for field-expedient cooling however this method requires a large volume (~40 gallons/151 Liters) of water, which may limit its practicality. Cold water dousing is a technique where patients are treated while lying supine on a porous stretcher resting on a tub filled with cold water (approximately 10-12°C). Medical personnel douse the patient with water and massage major muscle groups with ice bags without direct immersion in cold water. While cooling rates are somewhat lower than direct cold-water immersion, this technique allows continuous access to the patient, with a reported 100% survival rate (McDermott *et al.*, 2009b).

Another method is 'ice sheets', bed linens that have been soaked in ice water then applied to the entire body. The use of ice sheets has come under criticism, as available data suggests an inferior core temperature cooling rate (Butts *et al.*, 2017; Jardine *et al.*, 2020). However, the cooling rate reported in laboratory studies (Butts *et al.*, 2017) is limited by the magnitude of hyperthermia that is ethically permissible, typically no greater than 39.0-39.5°C. As the rate of heat transfer is largely dependent on the thermal gradient, the limited hyperthermia may under-estimate the effectiveness of the ice sheet methodology. Given that EHS casualties often have core temperatures >42°C, the core-to-skin thermal gradient and resultant heat transfer is likely greater as well. Military EHS casualties with an initial core temperature >39°C treated with ice sheets had a cooling rate of 0.16°C/min. (DeGroot, unpublished observations) Additionally, recent work evaluating the ice sheet methodology as currently employed by the US Army shows that the time to reach 38°C is significantly faster and the total thermal 'dose' (time * temperature area under the curve) is significantly reduced in the ice sheet trial vs. a passive cooling control condition, in which subjects rested in the heat, with no additional cooling (Caldwell *et al.*, 2022). In circumstances where cold water immersion is impractical, such as military field training away from fixed facilities, ice sheets should be considered as an acceptable alternative.

It may also be important to consider the cardiovascular and hemodynamic effects of cooling, whether via cold water immersion or ice sheets (Leon & Bouchama, 2015). Hypotension is commonly seen in hyperthermia and in EHS fatalities (Malamud *et al.*, 1946; Sawka *et al.*, 2011). Skin cooling will result in cutaneous vasoconstriction, improved venous return and central filling pressure and ultimately in reduced myocardial strain, partially off-setting the deleterious cardiovascular responses due to heat strain (Rowell, 1974). However, pre-hospital hemodynamic data in EHS casualties are lacking and the non-thermal benefits of rapid aggressive cooling remain to be elucidated.

Monitoring clinical markers (vital signs, skin color, and AVPU) while cooling an EHS victim can be used to track clinical improvement. If a rectal thermistor is used for initial temperature measurement, it can be left in place and the rectal temperature can be tracked during cooling. If a rectal thermometer is used for the initial body temperature measurement, repeat temperature measurements may not be needed to follow the clinical course during cooling as the markers outlined in Table 1 can effectively show improvements.

Whether using cold water immersion, ice sheets or some other methodology, the possibility exists that rapid cooling may result in shivering and agitation despite continued elevation of core temperature. Pharmacological intervention may be required to suppress shivering thermogenesis. While benzodiazepines are often considered for shivering, agitation and convulsions during cooling of EHS victims, there is little evidence-based guidance for specific therapeutic interventions (Roberts, 1992; Wexler, 2002). Clinicians should only use pharmacologic interventions which they are familiar with and ready to respond to any unintended consequences like respiratory suppression.

In the field setting cooling should only be stopped due to shivering only if core temperature is <39°C. If core temperature measurement is not available, cooling should be continued despite the presence of shivering. Under the most extreme stimuli, the maximum heat production due to shivering thermogenesis is ~5 times resting metabolic rate (Eyolfson *et al.*, 2001). However, shivering due to the application of ice sheets does not appear to affect the observed cooling rate.(Caldwell *et al.*, 2022) and it is better to have to re-warm a mildly hypothermic casualty than to stop cooling a suspected EHS casualty due to shivering.

When to stop cooling has not been well documented and it is probably safest to cool into the normal body temperature range. Stopping cooling at a rectal temperature in the 38-39ºC range has long been recommended but is not based on any scientific data. No specific threshold temperature for discontinuing cooling was determined in a range between 37 to 40°C body core temperature and there was no apparent difference in survival, but the associated long-term neurologic morbidity remains unknown (Bouchama *et al.*, 2007). However, based on the known susceptibility of the brain to extreme heat (Kiyatkin, 2010), it might be prudent to stop cooling at 37°C until robust data support discontinuing cooling at higher temperatures.

Data from rodent models allowed to cool passively into the hypothermia range during recovery from heatstroke, suggesting that low core body temperature may be protective for brain tissue (Leon *et al.*, 2005; Leon *et al.*, 2010). Extrapolating data from rodents to humans is not predictive and carefully designed clinical trials are needed before recommending a period of hypothermia post EHS. Using the markers in Table 1 as a clinical guide, cooling can be stopped when an EHS victim “wakes up” with a GCS nearing 15 and normal vital signs. Rectal temperature measurement at this point is desirable, but not essential. While there is some concern about overcooling, EHS casualties who “wake up” would rarely be cooled below 35ºC, which is the start of the hypothermic temperature range. After cooling, EHS casualties should be transferred to an emergency department for additional evaluation and observation. At some events where EHS is common, casualties may be observed in the medical area and discharged to home with follow up in 1-3 days with their primary physician to assess liver and kidney function for recovery.

Emergency Medical Transport

The emergency medical system (EMS) should be activated when EHS is suspected and onsite cooling should be started as soon as possible. If there is an onsite medical team, cooling should continue until the body temperature is lowered into a safe range and the medical care can be transferred to an advanced medical care facility. Algorithms should address onsite cooling strategies, indications for hydration, and importantly, criteria for patient hand-off from one level of care to the next. Coordination between the onsite medical care team, EMS facilitated transport, and the emergency department (ED) is critical and should be preplanned so communication and coordination of care with the ED allows the staff to prepare for the arrival of a heat casualty and expedite cooling.

The effectiveness of onsite cooling prior to transport to a higher level of care is clearly established by current evidence. Transporting an EHS casualty with a core temperature elevated above 39°C should only occur if en route cooling is available. EMS vehicles in areas with a high heat illness incidence rate should be equipped to begin or continue cooling therapy treatment while en route (chilled IV fluids, ice packs, cooling blankets [Bair Hugger™], fans) and use the vehicle air conditioning at high settings when EHS is suspected. Applying and rotating cold, wet towels may be feasible during medical transport (Belval *et al.*, 2018). Many EMS vehicles now carry refrigerated intravenous (IV) fluid chilled to 4˚C (39˚F) to augment induction of therapeutic hypothermia in cardiovascular emergencies (Bernard *et al.*, 2016). Cold saline infusion alone does not provide sufficient cooling in cardiovascular emergencies, however, when used in combination with other cooling strategies is associated with improved clinical outcomes and reduced hospital length of stay (Sinclair *et al.*, 2009; Mok *et al.*, 2017).

Advanced Clinical Care

The final phase of EHS clinical management involves advanced care in an ED and hospital with an inpatient critical care capability. This evidence-based review is not intended to outline either detailed ED, inpatient, or intensive clinical care management; there are multiple resources with detailed care protocols where key elements of management are discussed (Epstein & Yanovich, 2019; Liu *et al.*, 2020; Rublee *et al.*, 2021).

Ideally an individual with suspected EHS should be triaged and transported to a facility with experience treating heat casualties; EMS dispatch should notify the ED medical team in advance to allow preparation for immediate treatment upon arrival. The receiving facility should be equipped with whole-body cooling options and be prepared for an agitated and potentially combative patient who may require sedation, and emergency advanced airway management. The medical team must have the capacity to diagnose and manage common EHS sequelae including acute kidney injury, rhabdomyolysis, liver failure, and disseminated intravascular coagulation. Advanced heat stroke management may necessitate a facility with capabilities to include renal replacement therapy, extracorporeal membrane oxygenation (ECMO) and potentially early liver transplantation.

The critical interventions in the ED include heat stroke recognition, patient cooling and support care (Rublee *et al.*, 2021). A variety of new cooling devices have been developed in response to recent findings that cold-induced hypothermia may protect the brain following cardiac arrest including cooling catheters, helmets, pads, and blankets (Bernard *et al.*, 2002). The efficacy of targeted temperature management for patients post cardiac arrest has recently been questioned (Granfeldt *et al.*, 2021). However, the efficacy of these devices should be studied in EHS patients to assess where they may fit in cooling algorithms. The choice of cooling technique should be based on the availability of cooling devices and healthcare providers' familiarity with their use. (Bouchama *et al.*, 2007). No evidence-based adjuvant pharmacological agents have been shown to accelerate cooling, including dantrolene sodium or antipyretics (Bouchama *et al.*, 1991; Leon & Bouchama, 2015).

Supportive care is often required while cooling an EHS patient (Bouchama *et al.*, 2022). Basic and advanced cardiac life support, comparable to those used in any emergency, may be required in EHS, but should not interfere with the cooling process. Decision making for emergency airway intubation and mechanical ventilation should be based on clinical findings, as arterial blood gas results are difficult to interpret in hyperthermic conditions with significant acid-base disturbances (Bouchama & De Vol, 2001). Hypotension can be effectively treated with volume replacement during cooling and cooling will also return volume to the circulation. Invasive monitoring may be required when clinically indicated (Bouchama *et al.*, 2007). The management of heatstroke complications such as acute respiratory distress syndrome, acute kidney injury, disseminated intravascular coagulopathy, liver dysfunction, seizures, and coma require the same organ support provided for other critically ill patients. Figure 3 is adapted from a recently published evidence-based approach to EHS advanced clinical management in the ED (Rublee *et al.*, 2021). This algorithm assumes that no pre-hospital cooling has been completed and may not be fully applicable in other circumstances.

Insert Figure 3 about here

Recovery and Return to Activity

Recovery from EHS and return to activity (RTA) is usually straightforward and unremarkable provided adequate time is allowed for body healing (O'Connor *et al.*, 2018; Roberts *et al.*, 2021; Bouchama *et al.*, 2022). While most appropriately treated EHS casualties recover with full RTA, more serious EHS cases have demonstrated long-term complications including multi-system organ (liver, kidney, muscle) dysfunction, neurologic damage, reduced exercise capacity, and/or heat intolerance (Mehta & Baker, 1970). The duration of body temperature elevation is directly related to the prognosis with less time above the critical body temperature resulting in fewer difficulties in RTA. (Vicario *et al.*, 1986; Liu *et al.*, 2020). Studies analyzing the long-term quality of life following exertional or classic heatstroke are limited (Bouchama & Knochel, 2002; Leon & Bouchama, 2015). Case reports have identified cerebellar damage characterized by diminished Purkinje cells and atrophy of the cerebellum resulting in severe disability (Li *et al.*, 2015; Cheshire, 2016). In addition to the cerebellar damage, imaging studies performed years after heatstroke have demonstrated impairment to other regions of the brain, including the hippocampus, midbrain, and thalamus (Bazille *et al.*, 2005; Ookura *et al.*, 2009; Guivarch *et al.*, 2012; Fuse *et al.*, 2013; Guerrero *et al.*, 2013).

With the emergence of severe heat waves recent studies are systematically examining the long-term consequences of classic heatstroke, but similar studies are limited for EHS casualties. Classic heat stroke two-year follow-up data shows that up to 28% of patients who survive sustain cognitive and functional neurologic damage that persists unchanged. (Dematte *et al.*, 1998; Argaud *et al.*, 2007; Leon & Bouchama, 2015). In EHS, 11% of victims from 17 years of the Falmouth Road Race sustained a second EHS episode within the next 2 years, suggesting a period of increased risk even in promptly recognized and cooled EHS victims (Stearns *et al.*, 2020). Studies of French and US military personnel show an increase in risk for recurrence of EHS following a previous episode (Phinney *et al.*, 2001; Abriat *et al.*, 2014). Nelson et al found that active-duty soldiers with a prior serious exertional heat injury (EHI) had 4-fold increase in mild EHI events and those with a prior mild EHI had a 1.8-fold greater risk of having a serious EHI event (Nelson *et al.*, 2018). Finally, military personnel hospitalized for severe heat illness had a 40% increased risk for future cause-specific and total mortality; males with severe heat illness had an increased rate of death from cardiovascular disease (rate ratio (RR)=1.71, 95% confidence interval (CI): 1.01, 2.89) and ischemic heart disease (IHD) (RR=2.23, 95% CI: 1.02, 4.90) compared with reference cases (Wallace *et al.*, 2007).

Evidence-based data to guide RTA following EHS is limited. Current research suggests that most individuals recover completely within a few weeks, if the EHS event was promptly identified and cooled aggressively (McDermott *et al.*, 2007; Laitano *et al.*, 2019). Current RTA strategies and algorithms are largely based on anecdotal observations and consensus recommendations, relying on provider experience for decision making to safely advance activity (Armstrong *et al.*, 2007; O'Connor *et al.*, 2007; Asplund & O'Connor, 2016; Roberts *et al.*, 2021). Most RTA guidelines require the patient be asymptomatic at rest with normal end organ function (e.g., liver and kidney) before starting a cautious reintroduction of physical activity and a gradual heat acclimatization program. While there is wide variability in the recovery of affected organs, recent research does provide general guidance on biomarker recovery (Stearns *et al.*, 2016; Liu *et al.*, 2020). A retrospective review of 2,529 EHS episodes demonstrated that acute kidney injury (elevated creatinine, blood urea nitrogen, and blood and protein in urine) generally peak and normalize within 24-48 hours of injury; muscle damage and liver function-associated markers tend to peak within four days after injury and can persist outside the reference ranges for 2-16 days; and biochemical recovery from EHS is complete within 16 days for most casualties (Ward *et al.*, 2020). Assessing the basic hematologic parameters and blood chemistries for normal renal, hepatic, and coagulation function will give a baseline at the onset of activity, however, validated instruments to accurately predict continuing tissue or organ damage are currently not available.

A progressive increase in exercise intensity and duration in warm and hot conditions can be used to assess continued clinical recovery. If an individual can gradually regain exercise tolerance under heat stress conditions, RTA is usually safe and gradually advancing to full training and competition can be permitted. Research is needed to evaluate the role of a graduated increase in structured physical exercise to assist in return to activity decisions (McDermott *et al.*, 2007). Table 3 outlines a staged exercise and environment acclimatization return to activity program after EHS that incorporates the concepts of medical recovery (Roberts *et al.*, 2021).

Insert Table 3 about here

Heat tolerance testing (HTT) is a clinical tool that may be considered for individuals who are unable to advance activities in a reasonable time frame or who have had recurrent episodes of exertional heat illness. The Israel Defense Force (IDF) Medical Corps uses an exercise HTT about 6 weeks post-exertional heat illness as part of the “return to duty” criteria based on changes in rectal temperature and heart rate (HR) during the test (Moran *et al.*, 2004). A recent study of IDF military personnel calculated the sensitivity, specificity, and diagnostic accuracy of the HTT at 66.7%, 77.7%, and 77.2%, respectively with the authors concluding that the risk of heat stroke recurrence is measurable and that a negative HTT result is associated with a substantial reduction of future heat stroke risk (Schermann *et al.*, 2018). While case studies have demonstrated the efficacy of the IDF HTT in determining return to activity for military personnel and athletes afflicted by EHS there are concerns about the test's internal and external validity (Moran *et al.*, 2007; Roberts *et al.*, 2016). In a review article, Mitchell et al question both the test's external validity for tasks with high metabolic load, and the ability of the test to accurately determine true recovery and predict people who are not likely to experience another EHS episode (Mitchell *et al.*, 2019). Subsequent work from the same group revealed that heat acclimation is positively related to the pass rate and that the pass rate is dependent on the pass/fail criteria utilized. (Mitchell *et al.*, 2021) The role of the HTT in EHS return to activity is not fully understood.

Future Directions

This review discusses all aspects of EHS treatment and recovery, from initial recognition through return-to-activity decision making. While much is known about the care of the EHS casualty, numerous gaps remain. The ability to identify an EHS casualty prior to collapse is an area of active investigation. Once cooling measures have been initiated, there is no consensus regarding the core temperature at which active cooling should stop. The cardiovascular and hemodynamic effects of cooling, and subsequent effects on morbidity and mortality are unknown and future investigation in warranted. Biomarkers to predict the severity of co-morbid condition are lacking. The return-to-play decision making process has been developed from best practices and is not supported by outcomes data. The threat of greater heat stress due to climate change poses an ever-increasing risk to athletes, workers, recreational exercisers and members of the military, and will require continued research to address the gaps in EHS care.

Disclaimer: The views presented in this paper are those of the authors and do not necessarily represent the views of the Department of the Army, the Defense Health Agency, Department of Defense or the U.S. government.

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Figure Legends.

Figure 1. Chain of survival for an exertional heat stroke casualty.

Figure 2. Rectal temperature vs time hypothetical cooling curve where red arrow is time of collapse, orange arrow is time for evaluation & recognition, yellow arrow is start of cooling, and green arrow is end of cooling. The rectangular area in degree minutes is a function of evaluation time and highly dependent on the time between collapse and the start of cooling. If the time to recognition and cooling is 60 minutes rather than 10 minutes, the time in the "death zone" (rectangle) is markedly increased. The area in degree minutes once cooling has started forms a triangle that is influenced by the cooling rate (slope of the curve between yellow and green arrows). Cooling to the 38-39ºC range within 20 to 40 minutes usually produces good clinical outcomes. Cooling rates are usually increased later in the course of cooling.

Figure 3. Evidence-based management of heatstroke in the emergency department. CBC, complete blood count; CMP, comprehensive metabolic panel; CK, creatine kinase; PT/INR, prothrombin time/international normalized ratio; ICU, intensive care unit. Adapted with permission from (Rublee *et al.*, 2021)

Table 1: Monitoring Clinical Markers of EHS. Clinicians are cautioned that characteristics of each AVPU category are not fixed and subject to individual variability.

Clinical Marker	← Clinical Status Improving			
AVPU	**A**lert (GCS 15)	**V**erbal (GCS 14 to 9)	**P**ain Stimulus Response (GCS 8 to 4)	**U**nconscious (GCS 4 to 0)
Eye Opening	Spontaneous	Speech	Pain	None
Verbal Response	Oriented person, place, & time	Confused	Inappropriate (like echolalia) or incomprehensible	No response
Muscle tone	Normal	Protect face		Flaccid
Motor response	Normal	Withdraw	Abnormal flexion or extension	No response
Behavior	Normal		Personality change or aggressive behavior	No response
Facial skin color	Normal	Flushed		Pale, ashen
Heart rate	Normal sinus rhythm			Tachycardic
Systolic BP	Normotensive			Hypotensive
Rectal Temp	Normal	Mildly hyperthermic		Profoundly hyperthermic

Table 2: Common onsite whole-body cooling strategies for EHS casualties, with cooling rates as reported in published studies. Original research references provide detailed treatment implementation recommendations. Cooling rate ranges are affected by body type, initial body temperature (mild exercise-induced hyperthermia or heat stroke), and environmental conditions. Reproduced with permission from (Roberts *et al.*, 2021)

Body Cooling Strategies	Treatment Notes	Approximate Cooling Rate (°C/min)
Ice water (~2 °C) or cold water (~20 °C) immersion with stirring	Immerse the full body up to the neck including upper and lower extremities (~90% body surface area) in a tank/tub, circulate the water to increase heat transfer, add ice during cooling to maintain water temperature, support head and airway above water level.	0.13-0.35
	Immerse as much of the torso and pelvic region (~65% body surface) following full body treatment notes, extremities not in water should be cooled using other strategies	0.04-0.25
Cold water dousing	Free flowing hose or bucket with cool tap water applied to the whole body- extremities, torso, hands, feet, neck, and head (with attention to the airway).	0.04- 0.20
Tarp-assisted water ice/cold immersion	Providers hold the sides of the tarp with patient, water, and ice in the middle. Ensure as much of the torso, groin, and extremities are immersed as possible. Circulate the water as able.	0.14-0.17
Ice/cold water-soaked towels	Towels should be applied to the limbs (including feet and hands), trunk, and head with ice packs placed in the groin, axilla, and neck; include as much of the body as possible (~90% body surface area). Wring towels after soaking in bucket of ice water and change the towels rapidly.	0.11-0.16
Ice/cold water-soaked sheets	The whole body (~90% of body surface area) is wrapped in large sheets that are soaked with cold water. Sheets stay in place and are frequently rewetted. A fan directed at the body can be added.	0.05-0.06
Cold water immersion in portable water-impermeable bag	Immerse the full body up to the neck including upper and lower extremities (~90% body surface area) in the bag, circulate the water to increase heat transfer, add ice during cooling to maintain water temperature, support head and airway out of the bag.	0.04
Water spray/mister or high-powered fan with water spray	Patient should be placed supine on a cot or table. As much of the body surface should be exposed (i.e., remove clothes and shoes) to the fan and mist as possible.	0.03- 0.17

Table 3. Staged return to activity/play guidelines after exertional heat stroke. Reproduced with permission from (Roberts *et al.*, 2021)

Stage	Aim / Responsibility / Goal	Activity	Duration / Intensity	Example: HS Cross Country Runner
1	**Early Medical Recovery** Physician guided Organ system recovery	Activities of daily living for 1-2 weeks	Gradual increase in home activities without fatigue	Home rest & return to school
2	**Mid Medical Recovery** Physician Guided Sustain minimal aerobic fitness & develop confidence	Self-paced comfortable walk in low heat stress conditions (e.g., an air-conditioned gymnasium)	20-60 min at Maximal Intensity of HR<100 or <50% Age Adjusted maximal HR	Return to practice and walk through the warmup & practice, if the environmental conditions are not stressful. If not, use an air conditions area of the school
3	**Early Exercise Adaptation** Athletic trainer guided with physician Gradually improve aerobic exercise capability	Walk at 3.5 mph in low heat stress conditions	60 min at HR<140 bpm or <70% of age adjusted Maximal HR	Warm up & cool down with team, 1 of 4 reps at half speed
4	**Mid Exercise Adaptation** Athletic Trainer Guided with Physician Gradually improve aerobic exercise capability & fitness	Walk & run in low heat stress conditions	60 min of progressively increasing run to walk ratio until constant run for 60 min	1 of 3 reps, half speed 1 of 2 reps, ¾ speed
5	**Heat Acclimatization** Athletic Trainer Guided Gradually improve heat acclimation status	Run in ambient warm or hot conditions	60 min of progressively increasing run until constant run for 60 min;	All reps, ¾ speed
6	**Sports Specific Acclimatization / Training** Athletic trainer and/or coach guided Improve sport specific heat acclimation & fitness	Participate in practice in ambient conditions	Initially participate in sports specific drills with sports specific equipment then progress to training and scrimmage.	All reps, full speed
7	**Full Return to Sport** Athletic trainer monitors during warm up and game	Normal game or competition participation in ambient conditions		Meet 1 run to finish the race Meet 2 race to place

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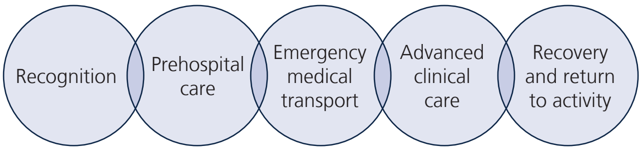
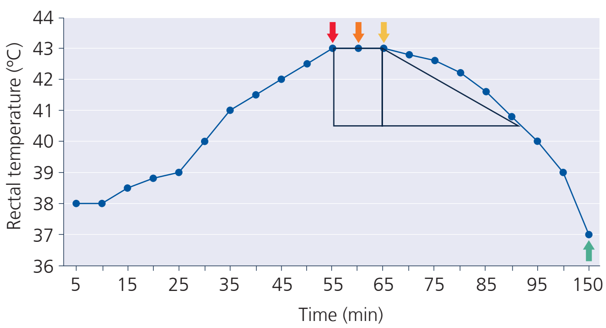


Experimental Physiology



Rectal temperature (°C)
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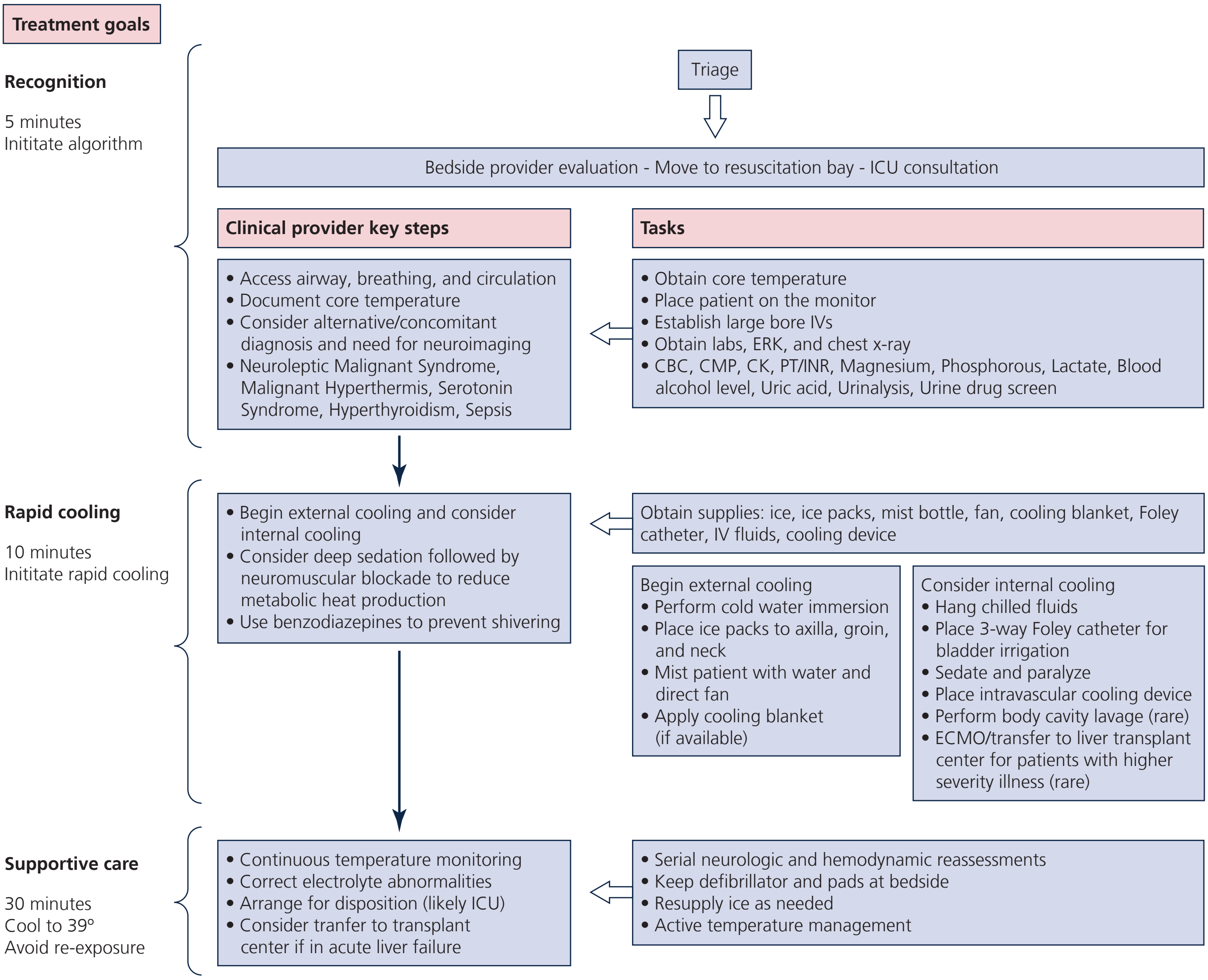

Experimental Physiology